

Image Classification: Cats vs Dogs

AI Internship Final Capstone Project · Ahmed Tareq

Problem Definition

This project builds a binary image classifier that distinguishes photographs of cats from photographs of dogs. Given the volume and variety of real-world images (different breeds, poses, lighting, and backgrounds), the goal is a Convolutional Neural Network (CNN) that generalizes to new, unseen photos rather than one that only performs well on images it has already seen — making this a supervised binary image classification task, evaluated on both held-out validation data and manually sourced test photos.

Dataset Description

The dataset is the Kaggle Cats vs Dogs collection, comprising 24,991 labeled images across two classes (Cat, Dog) after removing a small number of corrupted or unreadable files during preprocessing. Images were split 80/20 into 19,993 training and 4,998 validation examples, resized to 128×128 pixels, and pixel values rescaled from [0, 255] to [0, 1] to match standard CNN input expectations.

Model(s) Used

A CNN was built with three convolutional blocks (Conv2D with 32, 64, and 128 filters respectively, each followed by MaxPooling), flattened into a Dense(128, ReLU) layer, and a final Dense(1, sigmoid) output for binary classification. The model was compiled with binary cross-entropy loss and the Adam optimizer.

An initial baseline version was trained for a fixed 10 epochs with no regularization. After that version showed clear overfitting, three fixes were applied and retrained as the final model:

- Data augmentation (random horizontal flip, rotation, zoom) applied as a model layer, active only during training.
- A Dropout(0.5) layer before the output layer, to reduce co-adaptation between neurons.
- EarlyStopping (monitoring validation loss, patience of 3 epochs, restoring the best-performing weights) in place of a fixed epoch count.

Results & Analysis

Metric	Baseline (no regularization)	Regularized (final)
Training stopped at	Epoch 10 (fixed)	Epoch 13 (early stopping)
Training accuracy	98.7%	~85.7%
Validation accuracy	82.3%	86.7%
Validation loss (best)	0.91 (rising)	0.306 (stable)
Train/val gap	~16 points, widening	~1 point, stable

The baseline model reached 82.3% validation accuracy, but its training accuracy climbed to 98.7% while validation loss more than doubled over the same epochs — a clear overfitting signature. After applying augmentation, dropout, and early stopping, the retrained model achieved a higher 86.7% validation accuracy with training and validation

performance tracking closely together, indicating a model that generalizes rather than memorizes. Manual testing on two freshly downloaded photos (not part of the dataset) confirmed this: a cat photo was correctly classified at 80.7% confidence and a dog photo at 99.2% confidence — notably, the baseline model's confidence on the same cat photo had been an artificially inflated 99.999%, itself a symptom of overfitting rather than genuine certainty.

Future Improvements

- Use a truly held-out third test set, separate from the validation set used for early stopping, for a more rigorous final accuracy claim.
- Apply transfer learning with a pretrained backbone (e.g., MobileNetV2), which would likely raise accuracy further with less training time given the dataset's moderate size.
- Add a confusion matrix and per-class precision/recall to identify whether misclassifications skew toward one class over the other.
- Deploy the trained model as a public live demo via Streamlit, allowing anyone to upload a photo and receive a real-time prediction.